

Cordless technology in a mobile environment

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Despite the continued increase in demand for mobile cellular services there is a significant market opportunity for the second generation cordless systems, e.g. DECT and CT2. When the DTI allows UK operators to deploy wireless access systems based on cordless technology (not before at least 1998), BT may be given the opportunity to use such systems to replace final drops which are costly to install or maintain. Meanwhile there are significant opportunities for access cost reduction and service enhancement by deploying cordless installations as items of customer premises equipment. Cordless services in the home and office can be enhanced by using the fixed network to provide roaming between the home and office domains. The network would register the presence of the users at their homes or offices and use this information to direct their incoming calls to their present location. DECT roaming could be extended to a 'OnePhone service', based on DECT/GSM dual-mode handsets, which will provide a GSM service when the customer is out of range of DECT installations. OnePhone complements a GSM service and, for the fixed network operator, it minimises the traffic which is lost to mobile networks because of their attractions of contactability and a single number. This paper outlines the possible architecture and service options, reviews the status of the DECT standard and summarises the present product range.

1. Introduction

Two second generation cordless telephone standards have been developed in Europe for domestic and office applications, as well as for public telepoint services and radio local loop. The first of the standards was CT2. This was followed by the ETSI Digital Enhanced Cordless Telecommunications (DECT) standard which addressed the same markets while offering a higher capacity and greater functionality. Other cordless standards, which combine some of the technical features of both CT2 and DECT, have been developed in Japan (Personal Handy System — PHS) and in the USA (Personal Access Communications System — PACS).

Despite the continued increase in demand for mobile cellular services it will be argued that there will be a significant market opportunity for the second generation cordless systems. BT's opportunities for deploying DECT in the UK will be emphasised, but the architectures and services described are also applicable to the other standards. Which cordless system will be used in a given country, and for a given application, will depend on spectrum allocation and traffic levels.

Radio local loop systems based on second generation digital cordless technology would allow BT to offer flexible and rapid provision of service within about 100 m of a network distribution point. Applications of such a 'wireless access' system include the replacement of low drop wires, the provision of service to high rise flats and a number of other niche areas where wireless access can reduce the

whole life cost of the final drop. Regrettably, the DTI have indicated that they will not allow any operator to deploy wireless access systems in the UK until at least 1998. BT's plans for the deployment of wireless access have thus been re-focused on applications in which the radio base-station could be provided as an item of customer premises equipment (CPE). There are no regulatory constraints on such a service since the customer would be operating the radio base-stations under their Self-Provision Licence (SPL) and service could be offered directly to a handset or to a fixed radio unit. Such installations can:

- reduce fixed network costs and provide enhanced services and revenues on self-contained sites — 'site wireless access',
- complement mobile services by providing DECT roaming between offices and between offices and home — 'work and home wireless access'. Ubiquitous mobility could be provided by using a DECT/GSM dual-mode handset. This 'OnePhone' service would provide PSTN-quality speech and flexible data rates at or near-PSTN tariffs at the home and office and GSM services elsewhere. In addition to enhancing a GSM service, DECT roaming will help minimise mobile substitution of fixed network traffic.

Either CT2 or DECT could be used in some of these applications. Both are small-cell technologies (<100 m) providing 32 kbit/s ADPCM speech, but DECT is likely to

be preferred for most wireless access applications on the grounds of:

- flexible data rates ($n \times 32$ kbit/s),
- encryption and seamless handover,
- high traffic-handling capacity (future proofed),
- potentially lower costs because of larger volume.

A trial of a DECT-based service was successfully completed in January 1996 at a BT office complex in Colchester. There are now discussions with manufacturers to define the migration path to roll out systems, and a marketing trial of the DECT/GSM OnePhone services is currently being planned.

This paper will outline the possible architecture and service options, review the status of the DECT standard, summarise the present product range, and describe the possible service offerings.

2. Architecture and service options

The wireless access system which cannot be deployed under the present regulatory regime is depicted in Fig 1. In such systems a street mounted radio distribution point (DP) provides a link either to a handset or to a fixed radio unit capable of supporting the standard range of CPE.

Wireless access systems which can be deployed under the present regulatory environment are shown in Fig 2. In all of these, DECT equipment is connected to a standard network terminating point (NTP) on the customer's premises. These installations are domestic and small office / home office systems, wireless PBXs and, finally, reduced-

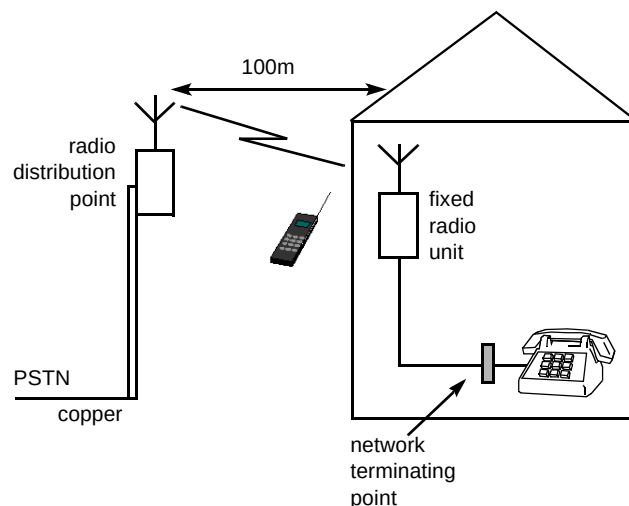


Fig 1 Radio local loop architecture.

functionality radio DPs connected to exchange-based controllers.

For large office applications, and for self-contained sites, it is proposed that external radio DPs be used wherever possible. This can drastically reduce the initial cost of providing a wireless infrastructure and thereby remove a significant barrier to unlocking the demand for cordless services in the office. Viability of the basic concept of using external radio DPs to provide services to multi-storey buildings has been demonstrated by the Colchester trial, and there is now a range of equipment that could be adapted to provide a site wireless access service. There are a number of applications where such a service would be of benefit to both BT and the user:

- provision of cordless and fixed services on self-contained sites, e.g. university campuses, hospitals and marinas, etc — cashless calling would be used for

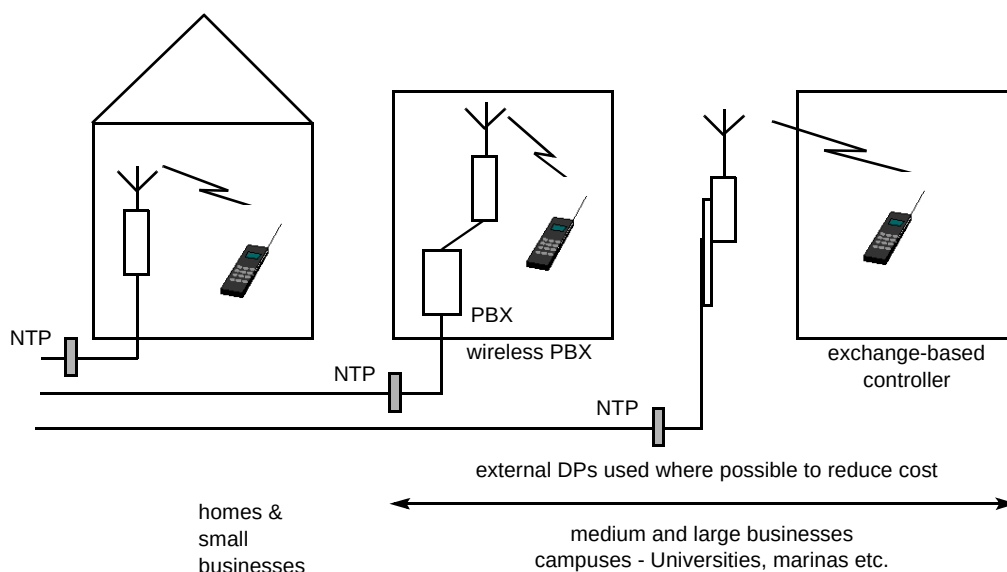


Fig 2 Deployment options for wireless access under the present regulatory constraints.

outgoing calls for students, patients and visitors, while the charging for employees would be as for a PBX, the wireless access system being owned or rented by the site owner,

- reducing the provisioning and maintenance costs of access infrastructure in commercial buildings, the cost reductions being achieved by the concentrating function of the radio DP and by using externally mounted radio DPs wherever possible,
- reducing the cost of providing existing centrex services by using the concentrating feature of radio DPs to minimise the number of additional copper pairs required.

The full range of DECT terminal options is illustrated in Fig 3.

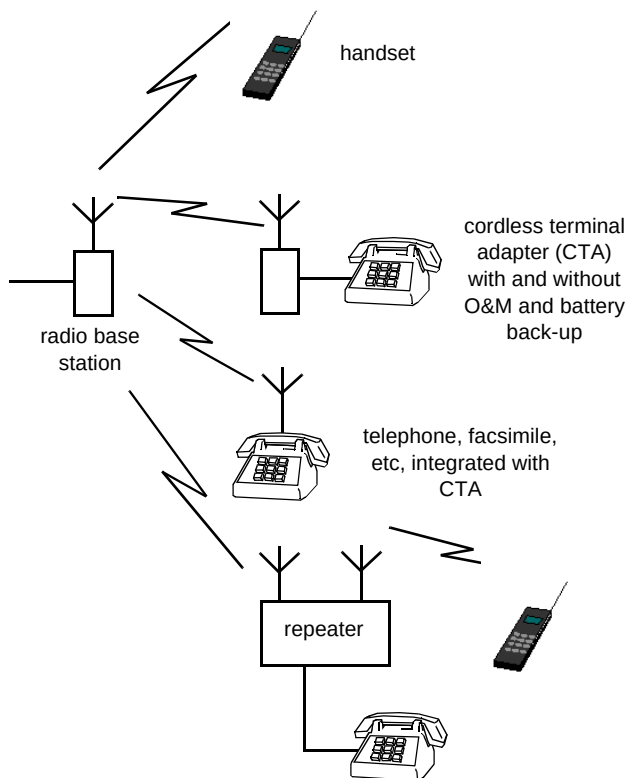


Fig 3 Cordless terminal options

Services can be provided to either handsets or to cordless terminal adapters (CTAs) which provide a cordless link to a standard phone or facsimile machine. CTAs can be integrated with a facsimile machine or telephone, and manufacturers are already working on DECT-based PCMCIA modem cards. They can be moved around the building and thus eliminate the need to change the access infrastructure in response to organisational changes, and they can have battery back-up and the provision for remote testing.

Repeaters are being produced to extend the range of the DECT system and to improve the in-building coverage. These repeaters have an NTP which will support fixed terminals and they are thus classed as DECT CPE for regulatory purposes.

Customer feedback from the wireless access trial in Colchester has provided ample evidence of the benefits of cordless communications in offices. Users were more confident because of their increased contactability and reported that they were working more efficiently. Thus, as the cost of wireless access systems fall, the take-up in offices is likely to rival that of cordless systems in the home (currently about 30%). Such high levels of cordless use could not be supported by the present generation of mobile systems. Neither could these mobile systems match the PSTN call charges, toll-quality speech and excellent in-building coverage of a digital cordless system. The new generation of micro GSM base stations, and improvements in the GSM speech CODEC, will go some way to overcoming these limitations; however, a significant difference between GSM and PSTN tariffs is likely to remain (at present it varies from about 2:1 to 3:1). Moreover, as DECT can already handle traffic densities well above that currently carried by wired networks in offices, a DECT-based office communications infrastructure is well equipped to handle future growth.

DECT cordless services in the home and the office can be enhanced by using the fixed network to provide roaming between the home and office domains — 'work and home wireless access'. The network would register the presence of the users at their homes or offices and use this information to direct their incoming calls to their present location. DECT roaming can be extended to a 'OnePhone service', based on DECT/GSM dual-mode handsets, which will provide a GSM service when the customer is out of range of DECT installations. Such a service complements a GSM service and, for the fixed network operator, it minimises the traffic lost because of the attractions of contactability and a single number.

A development version of a dual mode handset has already been produced by both Hagenuk and Ericsson. These and other companies have plans for production versions of the dual-mode handsets which are likely to be on the market after about mid-1997. Hagenuk were the first to have a development handset ready and they estimate that dual-mode functionality will add about 15% to the cost of a GSM phone.

DECT could also be used for public telepoint systems at railway stations, city centres and hotels, etc, but the experience of CT2-based telepoint systems suggests that such a service cannot compete with cellular services. It is difficult to cover large areas cost effectively with small cell technologies (such as DECT or CT2) when the traffic

density is low, because the base stations are not operating at their maximum capacity.

3. Status of the DECT standard

The DECT standard was produced by ETSI, and the European Commission has made it a mandatory standard for digital cordless systems throughout Europe. The standard provides a specification for the radio interface but not for the supporting network infrastructure, thus leaving the manufacturers flexibility in the implementation of DECT networks.

The DECT standard comprises basic standards and applications standards or profiles. The basic standards are the common interface and the approvals test specification which were approved by ETSI in 1992. The profiles ensure interoperability between units from different manufacturers, or between networks, for specific applications. The key profiles are:

- the public access profile (PAP) — defining the authentication procedures for public access (i.e. telepoint),
- the generic access profile (GAP) — allowing basic speech functionality to be supported between handsets and base stations from different manufacturers,
- data profiles — implementation of data services such as radio LANs,
- DECT/ISDN interworking profile — provision of ISDN services with DECT,
- DECT/GSM interworking profile (GIP) — describing how DECT can be connected to the GSM switching infrastructure.

For the applications to be discussed here, GAP is the most important profile. It will allow a handset to obtain basic speech services from a base station for all GAP-compliant manufacturers. GAP will become mandatory in the autumn of 1996 but several manufacturers have already demonstrated compliance.

Within buildings a DECT handset will have a range of about 100 m depending upon the construction of the walls and the position of the base stations. In an installation where coverage is provided over an extended area by overlapping cells, there is seamless handover as a user moves between cells.

For external base stations the range can be extended to about 300 m if the line-of-sight path to the base stations is not severely obstructed by buildings or trees. These larger cell sizes would be typical of a telepoint system. There have been claims that, in such applications, the range of DECT could be affected by the presence of multiple propagation

paths caused by reflections from surrounding buildings. Advocates of both the PHS and PACS rightly point out that their systems will be less susceptible to such propagation conditions than DECT. What is not clear, is whether the greater susceptibility of DECT to multiple paths will cause a problem in most applications and, if so, whether this will outweigh the service advantages that DECT has at present over the other systems.

DECT uses time division multiple access, time division duplex on ten channels in the frequency band 1.88 to 1.9 GHz. Dynamic channel allocation allows multiple users to share the same spectrum with the minimum of interference. In areas where there are multiple base stations having overlapping coverage areas, such as wireless PBXs or RLL installations, the transmit and receive cycles of these base stations need to be synchronised to minimise interference and thereby achieve the highest capacity.

Some potential operators in Europe want to use cells up to about 3 km in radius for local loop bypass. (Such installations are characterised by base stations with antennas above the roof line and by customer installations having external antennas which are also above the roof line. In such cases the service will be to a fixed telephone unless the customer's installation is in the form of a repeater as discussed in section 2.) Regulators are concerned that this may adversely affect domestic and office systems operating in the same band. A number of solutions are being considered:

- allocating additional spectrum,
- limiting RLL to specific channels in the DECT band,
- limiting the cell size for RLL.

In the UK, the Radiocommunications Agency have indicated that they will limit cell sizes to 200 m when they eventually licence an RLL service (not before 1998).

Recently the European Radiocommunications Office has entered the debate about how to mix local loop applications operated by licensed operators with the unlicensed small-cell installations in offices and homes. They suggest that unlicensed installations should be limited to internal use only. Such a rule would prevent owners of self-contained sites from benefiting from the significant cost savings associated with the use of external radio base stations. The rule could be challenged on the basis that if external base stations are below the roof line of the surrounding buildings, and if the antennas direct the radiated energy towards the buildings, the interference caused to other installations will not be significantly greater than that caused by internal base stations.

4. Current product range

A comprehensive range of DECT products, targeted at a number of applications, is beginning to emerge.

4.1 RLL systems for large sites

For large self-contained sites, e.g. university campuses or hospitals, where several hundred customers can share the cost of the infrastructure, the standard RLL product provides the most economic and flexible way of providing service. In principle, wireless PBXs could also be used for this application but the present product range is limited in the number and type of terminals that can be supported and does not include externally mounted radio DPs.

The RLL call controller is mounted on the site and is connected to the main network via a standard user interface, e.g. DASS2, as shown in Fig 4. Proprietary digital signalling over the copper access network is used to support both internally and externally mounted radio base stations. (It is worth noting that for some large-cell applications of DECT, the capacity of the base stations is such that it is economic to use fibre or microwave links to the call controller.)

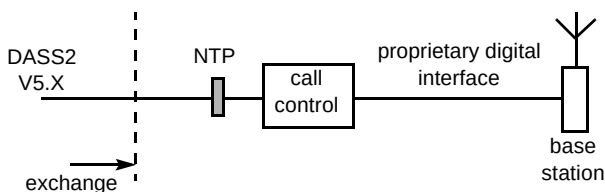


Fig 4 RLL system for large sites.

4.2 Modular systems for small to large sites

To provide a concise description of the product range which is suitable for these sites, attention will be focused on the two basic generic architectures. One has the call control in the network (i.e. at the local exchange) and the other has the call control at the customer's premises.

Centralised call control

Two forms of centralised call control have been considered:

- for the integrated call-control option of Fig 5(a), the proprietary digital interface of the RLL product is transformed into ISDN2, with the call control either being a stand-alone unit at the local exchange or absorbed into an embedded line card and the exchange software,

- for the adjunct call-control system of Fig 5(b), the radio base station sets up a standard network call to a remotely sited call-control unit whenever a request for service is made — once the link to the base station or the call control unit has been established it is used as a bit pipe and the system operates in a similar manner to RLL.

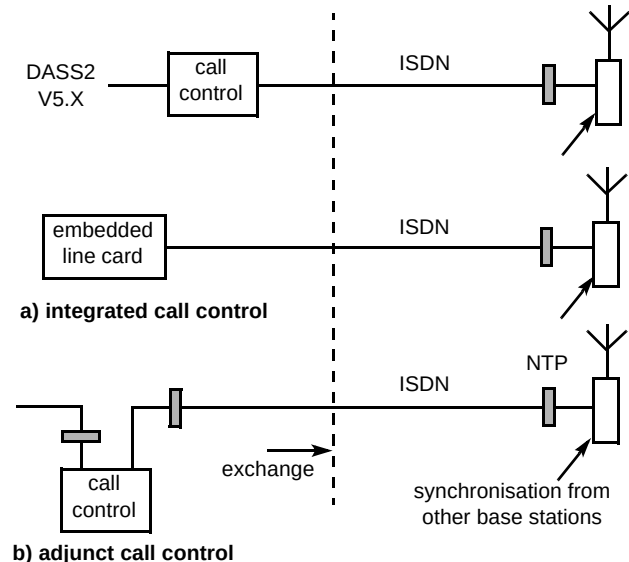


Fig 5 Centralised call control options.

Centralised call control is potentially the lowest cost solution for large and small businesses because the call-control function can be shared by many users. However, given that significant development work and investment would be required to introduce either of these systems into the network, they would not be practical until a market had been created by deploying the units discussed in the next section.

Call control at the customer's premises.

The generic architecture of systems having call control at the customer's premises is shown in Fig 6.

- The intelligent compact access node (ICAN) of Fig 6(a) can handle several base stations and can route intra-site calls without having to trombone them back to the main network. For wireless centrex applications the ICAN would have to offer service transparency and be compact. Both of these requirements have been discussed with manufacturers and neither would appear to present a problem.
- The compact access node (CAN) of Fig 6(b) is similar to the ICAN but it lacks service control or switching functions. Services are controlled by the supporting network and intra-site calls have to be tromboned via

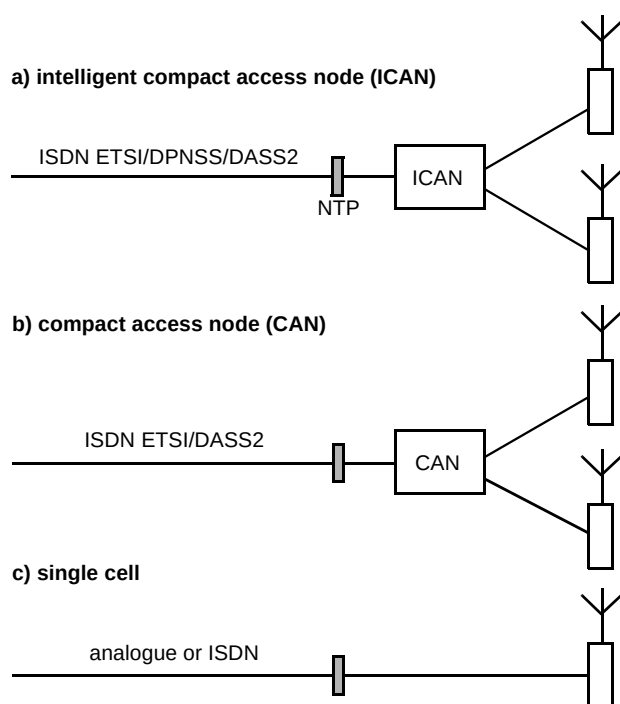


Fig 6 Call control at the customer's premises.

the supporting network. A CAN could be used for a wireless centrex service, or it could be added as an adjunct to a PBX to form a wireless PBX.

- Finally, as shown in Fig 6(c), there are units which can support multiple terminals in a single cell. These include domestic units and cordless 'key systems' in which the base station is integrated with a fixed telephone and is used to control the communications with the associated cordless terminals.

Driven primarily by the demand of the German market, there has been extensive development of ISDN-based ICANs, for small business and small office / home office applications. The development of CANs is in response to the market for cordless adjuncts to PBXs. Most ICANs and CANs are modular units, which can provide an economic solution for large and small installations.

CANs and ICANs both require either a 2-Mbit/s link to the local exchange (primary rate ISDN or DPNSS) or fractional primary rate. It is envisaged that the digital links for installations serving large- and medium-sized businesses would be based on high-speed digital subscriber loop (HDSL):

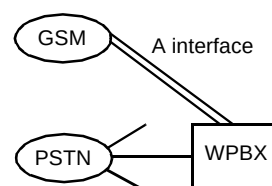
- a fully configured HDSL link would use three copper pairs and would feed a single CAN or ICAN module, typically supporting up to four radio DPs and providing a capacity of 19 Erlang,
- a sub-equipped HDSL link would use one copper pair and would provide a capacity of 4 Erlang.

The ICANs being developed for small office / home office applications require one or two ISDN2 links (providing two to four exchange lines respectively).

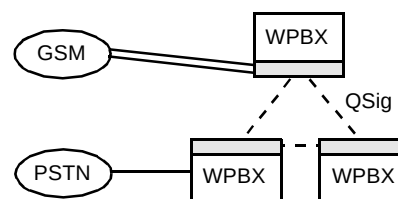
Both CANs or ICANs will allow the use of external DPs wherever possible. The links to these will be via twisted pairs and line powering will be possible for lengths up to about 2 km. Although both would be deployed as items of CPE, the present systems incorporate O&M functions which could be accessed by the core network.

5. DECT roaming

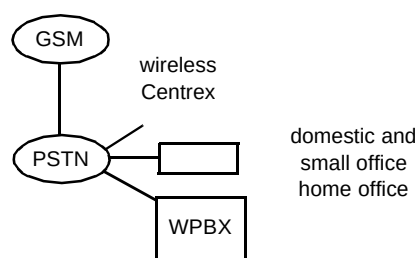
Groups within ETSI are currently defining standards for supporting DECT roaming. This work, under the banner 'cordless terminal mobility' (CTM), is based on the three architectures illustrated in Figs 7(a) — (c).



a) DECT roaming controlled by GSM network



b) DECT roaming controlled by WPBX private network



c) DECT roaming controlled by the PSTN

Fig 7 Architectures for implementing DECT roaming.

In Fig 7(a), each DECT installation has a GSM 'A' interface to a GSM switching network which provides both mobility management and call routing.

In Fig 7(b) (private networks of DECT installations), each installation is linked by Qsig (DPNSS) and this is used for mobility management and traffic.

In Fig 7(c), each DECT installation is linked to an IN using ISDN and an enhanced form of DSS1 (DSS1+) is used for mobility management.

The IN-based option is the only one on which it is practical and economic to support roaming across the full range of DECT installations, i.e. domestic, small office / home office, and all sizes of businesses. Accordingly, this is the option which is being pursued within BT.

Although the IN-based CTM solution would provide all the functionality required for both DECT roaming and DECT/GSM OnePhone services, it is not possible to predict when the BT network will support the ETSI CTM. There is a need, therefore, for a pre-IN CTM solution based on interworking boxes which will provide an interface between the existing network and CPE which has the necessary functionality to support CTM. These interworking boxes would be updated as the network develops.

Figure 8(a) illustrates the basic functionality of an interworking box for domestic users. It uses the network services 'call divert' function and the 'bypass number' service. The latter provides a main number which can be diverted and a bypass number which cannot.

Eventually, it would also be possible to use a distinctive ringing service to direct the call to a particular handset.

To provide roaming between sites at which the user was not pre-registered there would have to be a centralised authentication database as illustrated in Fig 8(b). This could be a stand-alone unit or it could be part of a personal number platform.

Prototype interworking boxes have been developed at BT Laboratories to stimulate manufacturers to produce pre-IN CTM-based domestic and office installations for trials towards the end of 1996.

These prototypes are based on network services, but it is envisaged that the roaming will ultimately be based on the personal number service OnePhone users would have fixed network numbers and calls would be diverted to their mobile network operator when they were not within range of any of the DECT installations at which they were registered to receive service.

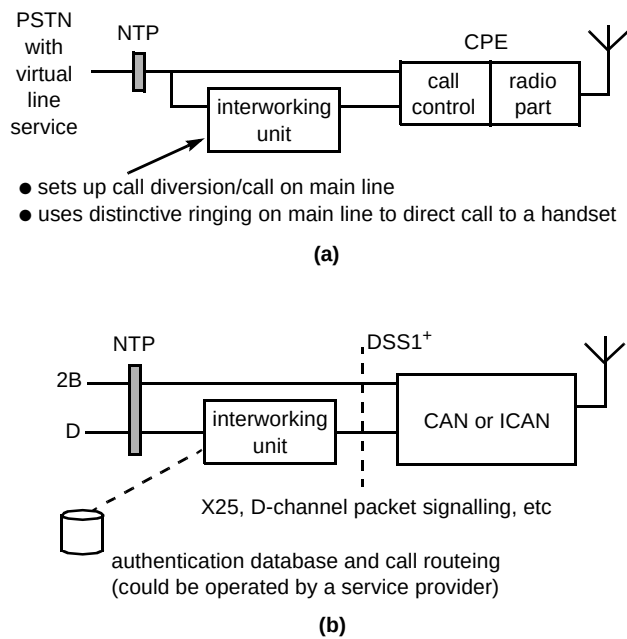


Fig 8 Examples of interworking boxes.

6. Service packages

Various service packages based upon DECT are currently being considered, and detailed cost models have been produced to support the associated market assessments. These service packages are outlined below. A detailed breakdown of the costs of the DECT-based services is beyond the scope of this review, but a comparison of the costs of DECT and GSM-based services are given in section 7.

6.1 Hospitals, university campuses and marinas

For large sites such as hospitals, campuses and marinas, BT has the opportunity to provide a turnkey, fully managed system. External radio DPs, making use of existing access infrastructure, would be installed where appropriate. The institution that owned or rented the installation would be responsible for distributing handsets to students, patients or visitors. All outgoing calls for these users would be via the BT cashless system. Calls made by staff would be billed to the institution. In typical installations, having several hundred handsets, the capital cost per user will be dominated by the cost of the handset. The retail price of these has fallen dramatically over the last year to about £100 and it is likely to reach the range £50—£70.

6.2 Small (>10 lines) to large businesses

The CAN and ICAN modular solution proposed for large and small sites provide a unified approach for:

- reducing the provisioning and maintenance costs of the internal access infrastructure in commercial buildings and the cost of providing centrex services,
- providing cordless services in offices,
- DECT roaming and DECT/GSM OnePhone services which will complement GSM services and reduce the loss of traffic from the fixed networks to the mobile networks.

BT could install the CANs, ICANs and base stations and, where appropriate, use existing access infrastructure to support external DPs. These external radio DPs would be the key to reducing the cost of wireless access installations which require multiple cells for coverage.

Although the CANs and ICANs would be marketed as items of CPE, as would the cordless terminal adapters and handsets, BT could offer a fully managed service which included these units.

The customer benefits, for which they will have to pay the cost of buying or renting the CPE, are:

- cordless mobility on site, and site-to-site roaming if they use the DECT roaming and OnePhone services, characterised by PSTN quality speech, flexible data rates, PSTN tariffs and good in-building coverage,
- freedom from the constraints of fixed wiring through the deployment of both handsets and cordless terminal adapters — the costs of CTAs will be comparable to that of handsets.

BT benefits in the following areas:

- reduction in capital and current account costs of the access network, these savings being achieved where BT has responsibility for the internal wiring in commercial premises, and by reducing the access infrastructure required to support centrex services — the saving in the access costs will vary from site to site and it is not yet clear whether these savings will cover the costs of the mobility and O&M features,
- increased revenue which will result from handset users becoming more contactable,
- reducing the loss of traffic to mobile networks from customers who want contactability and a single number.

6.3 Domestic and small business

The domestic and small business market which is based on single-cell systems is currently served by the BT Diverse 1000 unit which has one exchange line and can support up

to 6 handsets. The popularity of this new product is being attributed to the high speech quality and freedom from interference. Multiple-line ICAN systems for small office/home office applications will also soon be available in the UK. The addition of a pre-IN CTM interworking box to these units would provide a home base station for a DECT roaming or a DECT/GSM OnePhone service.

7. Cost comparison of DECT and GSM

As with any concentrating radio system, the costs per customer of DECT and GSM systems depend on traffic rates and on whether the base stations are operating at full capacity. The simplest way of comparing costs of the two systems is to calculate the cost per Erlang of a fully loaded system. This approach has been used to derive the data presented in Fig 9. Two sets of columns are given for both DECT and GSM. The left-hand column gives the present cost of the base stations and controllers and the right-hand column includes transmission and site costs.

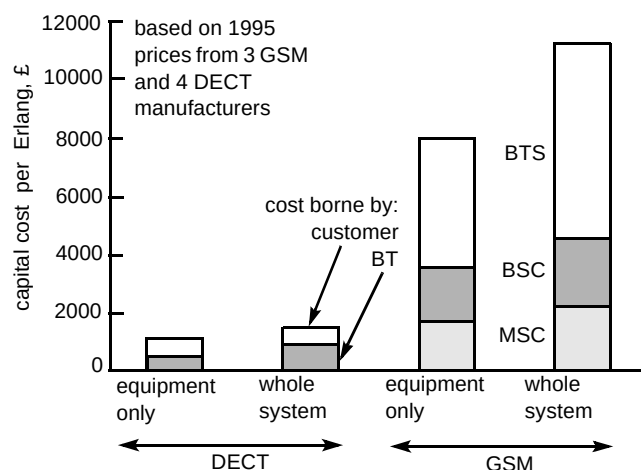


Fig 9 Comparison of DECT and GSM capital costs

For the GSM system the customer only pays for service, whereas for the DECT system the customer must rent or buy all but the transmission system which links the base station to the local exchange. In calculating the cost of the DECT transmission system the historic cost for the copper access network has been used.

In Fig 9 the segmentation of the costs for GSM shows the split between the base stations (BTS) and the various levels of switch (BSC and MSC). For DECT, the split is between the cost borne by BT and that by the customer. In neither case is the cost of the handset included. The cost per Erlang of DECT is significantly lower than that of GSM. It should be noted, however, that the costs are based on a fully loaded base station and that this is easier to achieve for a GSM system than for DECT. The costs for DECT are for an installation which would be typical of a large office. In such cases the fill factors for DECT could be as low as 50% in some installations and this will raise the cost per Erlang on a

pro rata basis. Nevertheless, the cost difference between DECT and GSM remains significant.

8. Conclusions

During the second quarter of 1996 there will be a wide variety of DECT equipment on the market for domestic, small home office and PBX applications. A number of manufacturers are about to submit handsets for GAP compliance approval and others have already demonstrated interworking of their products. There is a range of fixed radio adapters which will liberate offices from the constraints of internal wiring and there will soon be a number of prototype PCMCIA DECT data cards. DECT/GSM dual-mode handsets and DECT roaming services have received considerable attention and most of the major handset manufacturers have production plans for these during 1997/98. There are also a number of DECT local loop trials around the world and more are planned as a number of service scenarios are investigated. The cost of DECT equipment is also falling rapidly in response to competition and increased volumes. This is also true for both PACS and PHS. It would thus appear that second generation cordless telephony is about to blossom.

The opportunities that BT has for deploying DECT in the UK have been identified together with ways of implementing the required functionality. The next stage is to quantify fully the potential cost savings and service opportunities, implementing trials to refine the product offering and the whole life cost models.

A marketing trial of the DECT/GSM OnePhone service will be conducted by Network Independent Services. It will start in the late summer of 1996 and there is already a feasibility study in progress to assess the various architectural options which could be used to provide a service in about 1997.

A number of migration topics are also being addressed. These include both voice mail compatibility of the fixed and mobile networks, and service compatibility issues with GSM.

Acknowledgements

The help and support of Alan Chisholm, Tony Dunn, Paul Greenshaw, Adrian Pote, Louise Price, Mike Smith and Rob Warburton has been much appreciated.



Bob Merrett joined BT as an apprentice in 1965. After graduating from the University of Wales with a BSc in Electrical and Electronic Engineering in 1969 and a PhD from the University of Cambridge in 1972, he joined the Reliability Division at BT Laboratories where he led teams assessing the reliability of semiconductor packaging, reed relays and a range of discrete electronic devices. In 1983 he joined the Devices Division and led a team designing high-speed GaAs IC and then a team developing InP OEICs. Since 1990 he has worked on various aspects of radio local loop and currently manages the technical support work for DECT-based wireless access and the GSM/DECT OnePhone.



Steve Buttery joined BT in 1981 as an apprentice. Following this he worked on installation for two years before leaving for University as an award holder. Upon graduating, in 1989 he joined the radio division at BT Laboratories. Since then, he has worked on a number of different radio-related projects, ranging from writing software planning tools to supporting the development and testing of a rural radio access system. Currently, he is leading the technical work being carried out at BT Laboratories in support of the 'OnePhone' DECT/GSM project.